

1.A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

```
def encrypt(text, key):  
    result = ""  
    for ch in text:  
        if ch.isupper():  
            result += chr((ord(ch) - 65 + key) % 26 + 65)  
        elif ch.islower():  
            result += chr((ord(ch) - 97 + key) % 26 + 97)  
        else:  
            result += ch  
    return result
```

```
def decrypt(text, key):  
    result = ""  
    for ch in text:  
        if ch.isupper():  
            result += chr((ord(ch) - 65 - key) % 26 + 65)  
        elif ch.islower():  
            result += chr((ord(ch) - 97 - key) % 26 + 97)  
        else:  
            result += ch  
    return result
```

```
print("==== Caesar Cipher ====")  
print("1. Encrypt")  
print("2. Decrypt")
```

```
choice = input("Enter choice (1/2): ")
```

```
text = input("Enter text: ")  
key = int(input("Enter key: "))
```

```
if choice == "1":  
    print("Encrypted Text:", encrypt(text, key))  
elif choice == "2":  
    print("Decrypted Text:", decrypt(text, key))  
else:  
    print("Invalid Choice")
```

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OUTPUT:

```

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:63ae6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/info/Downloads/S073 cyber.py =====
----- Caesar Cipher -----
1. Encrypt
2. Decrypt
Enter choice (1/2): 1
Enter text: siddhant
Enter key: 4
Encrypted Text: vmhhlerx
>>>
===== RESTART: C:/Users/info/Downloads/S073 cyber.py =====
----- Caesar Cipher -----
1. Encrypt
2. Decrypt
Enter choice (1/2): 2
Enter text: siddhant
Enter key: 3
Decrypted Text: pfaser
>>>

```

```

from tkinter import *
from tkinter import messagebox

```

```

def encrypt(text, key):
    result = ""
    for ch in text:
        if ch.isupper():
            result += chr((ord(ch) - 65 + key) % 26 + 65)
        elif ch.islower():
            result += chr((ord(ch) - 97 + key) % 26 + 97)
        else:
            result += ch
    return result

```

```

def decrypt(text, key):
    result = ""
    for ch in text:
        if ch.isupper():
            result += chr((ord(ch) - 65 - key) % 26 + 65)
        elif ch.islower():
            result += chr((ord(ch) - 97 - key) % 26 + 97)
        else:
            result += ch

```

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```
return result
```

```
def do_encrypt():  
    try:  
        k = int(key.get())  
        result.set(encrypt(message.get(), k))  
    except ValueError:  
        messagebox.showerror("Error", "Please enter a valid numeric key.")
```

```
def do_decrypt():  
    try:  
        k = int(key.get())  
        result.set(decrypt(message.get(), k))  
    except ValueError:  
        messagebox.showerror("Error", "Please enter a valid numeric key.")
```

```
# Create window
```

```
root = Tk()  
root.title("Caesar Cipher")  
root.geometry("400x220")
```

```
# Variables
```

```
message = StringVar()  
key = StringVar()  
result = StringVar()
```

```
# Labels and Entries
```

```
Label(root, text="Message").grid(row=0, column=0, padx=10, pady=10, sticky="w")  
Entry(root, textvariable=message, width=30).grid(row=0, column=1)
```

```
Label(root, text="Key").grid(row=1, column=0, padx=10, pady=10, sticky="w")  
Entry(root, textvariable=key, width=30).grid(row=1, column=1)
```

```
# Buttons
```

```
Button(root, text="Encrypt", command=do_encrypt, width=12).grid(row=2, column=0, pady=10)  
Button(root, text="Decrypt", command=do_decrypt, width=12).grid(row=2, column=1, pady=10)
```

```
# Result
```

```
Label(root, text="Result").grid(row=3, column=0, padx=10, pady=10, sticky="w")  
Entry(root, textvariable=result, width=30, state="readonly").grid(row=3, column=1)
```

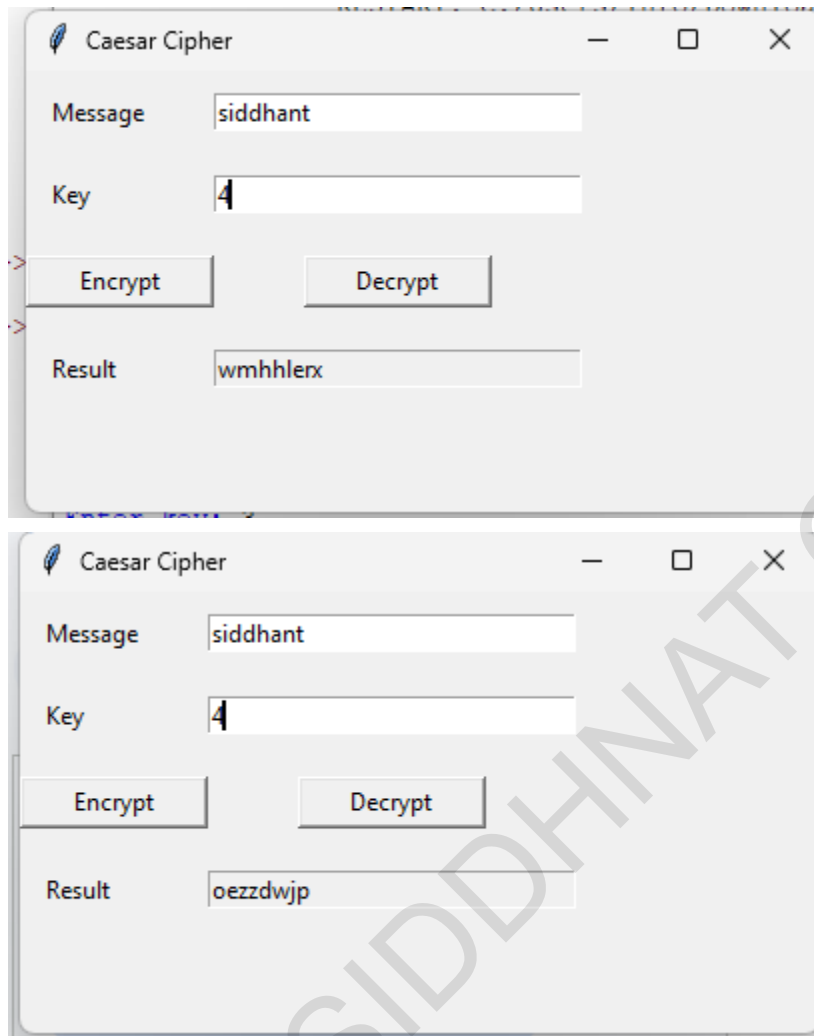
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root.mainloop()

OUTPUT:



1. B Design and implement algorithms to encrypt and decrypt messages using transposition technique

def encrypt(text, key):

 result = ""

 for ch in text:

 if ch.isalpha():

 base = ord('A') if ch.isupper() else ord('a')

 result += chr((ord(ch) - base + key) % 26 + base)

 else:

 result += ch

 return result

def decrypt(text, key):

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```

result = ""
for ch in text:
    if ch.isalpha():
        base = ord('A') if ch.isupper() else ord('a')
        result += chr((ord(ch) - base - key) % 26 + base)
    else:
        result += ch
return result

text = input("Enter message: ")
key = int(input("Enter key: "))

cipher = encrypt(text, key)
print("Encrypted:", cipher)

plain = decrypt(cipher, key)
print("Decrypted:", plain)

```

OUTPUT:



```

===== RESTART: C:/Users/info/Downloads/S073 cyber.py =====
Enter message: SIDHANT
Enter key: 3
Encrypted: VLGRDQW
Decrypted: SIDHANT
>>>

```

import tkinter as tk

```

def encrypt():
    text = msg.get()
    key = int(rail.get())

    rails = [" " for _ in range(key)]
    row = 0
    direction = 1

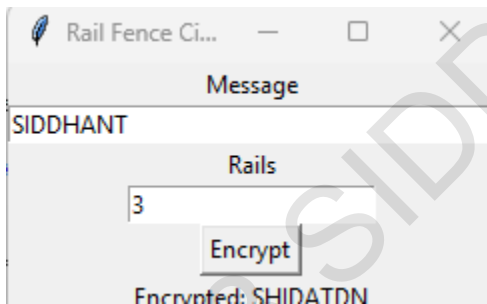
    for ch in text:
        rails[row] += ch
        row += direction

```

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```
if row == 0 or row == key-1:  
    direction *= -1  
  
output.config(text="Encrypted: " + ".join(rails))  
  
root = tk.Tk()  
root.title("Rail Fence Cipher")  
  
tk.Label(root, text="Message").pack()  
msg = tk.Entry(root, width=40)  
msg.pack()  
  
tk.Label(root, text="Rails").pack()  
rail = tk.Entry(root)  
rail.pack()  
  
tk.Button(root, text="Encrypt", command=encrypt).pack()  
  
output = tk.Label(root, text="")  
output.pack()  
  
root.mainloop()
```

OUTPUT:



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